4 **(a)** **(i)** With reference to the vibrations involved, state the difference between *transverse* and *longitudinal* waves.

.....

.....

.....

.....

..... [2]

(ii) State an example of each.

example of a transverse wave: .....

example of a longitudinal wave: ..... [1]

(b) **(i)** State what is meant by the diffraction of a wave.

……..

..... [1]

(ii) Explain how diffraction is essential for observable double-slit interference to occur. Sketch a diagram to help your answer to be clearer.

……..

……..

..... [2]

(c) Certain types of electromagnetic waves are used in radar systems where they are passed along tubes called waveguides. One part of the system can be a double tube as shown below, where the lengths of the wave paths in the tube are 18 cm and 31 cm as shown.

The wavelength of the radar waves in the waveguide is 4.0 cm.

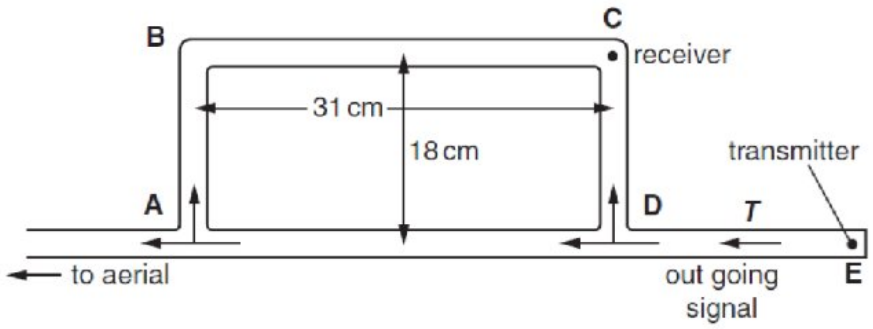


Fig. 4.1

(i) A wave ***T*** travelling from a transmitter at **E** in the direction shown in Fig. 4.1. It reaches a receiver at **C** by paths **DABC** and **DC**.

Calculate the path length of each of these paths.

path length **DC** = ... cm; path length **DABC** ..... cm [1]

(ii) Determine the phase difference between these two waves arriving at **C** and comment on the amplitude of the resultant wave at **C**.

phase difference = ... [2]

..... [1]

[Total: 10]